

GarageMinder Mobile API v1

Complete REST API for the GarageMinder mobile companion apps (Android & iOS). Coexists alongside the `/app/api.php` web app API.

Base URL

```
https://trackmywrench.com/api/v1
```

Authentication

JWT (JSON Web Tokens) via two login methods. All authenticated endpoints require:

```
Authorization: Bearer <access_token>
```

Access tokens expire in **30 minutes**. Refresh tokens expire in **30 days** and are rotated on each use.

Response Envelope

Every response uses this JSON structure:

```
{
  "success": true,
  "data": { ... },
  "error": null,
  "meta": {
    "api_version": "1.0.0",
    "timestamp": 1707580800
  }
}
```

Error responses:

```
{
  "success": false,
  "data": null,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Human-readable message",
    "details": { ... }
  },
  "meta": { ... }
}
```

Error Codes

400	BAD_REQUEST	Invalid input or malformed body
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401	UNAUTHORIZED / TOKEN_EXPIRED / INVALID_CREDENTIALS	Missing, invalid, or expired token
403	FORBIDDEN / ADMIN_REQUIRED	Insufficient permissions
404	NOT_FOUND	Resource does not exist or not owned by user
422	VALIDATION_ERROR / ODOMETER_DECREASE / ODOMETER_JUMP_TOO_LARGE	Field validation failed
429	RATE_LIMITED	Too many requests
500	INTERNAL_ERROR	Server error

Rate Limiting

Per IP: **200 req/min** · Per user: **100 req/min** · Login: **10 req/5 min**

Headers: X-RateLimit-Limit , X-RateLimit-Remaining , X-RateLimit-Reset

Important: ID Types

Vehicle and reminder IDs are **strings** (varchar 64), not integers. Example vehicle ID: `v_1766985747930_q4x1fvngxzde` .
Always treat IDs as opaque strings.



Authentication

POST /auth/login

Authenticate with username/email and password. Returns JWT access + refresh tokens.



No authentication required

REQUEST BODY

```
{
  "username": "user@example.com",
  "password": "yourpassword",
  "device_id": "android_abc123",
  "device_name": "Pixel 8 Pro",
  "platform": "android"
}
```

FIELD	TYPE	REQUIRED	DESCRIPTION
-------	------	----------	-------------

username	string	Yes	Username or email address
password	string	Yes	Account password
device_id	string	No	Unique device identifier for session tracking
device_name	string	No	Human-friendly device name
platform	string	No	android or ios

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": {
    "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "refresh_token": "a1b2c3d4e5f6g7h8i9j0...",
    "token_type": "Bearer",
    "expires_in": 1800,
    "user": {
      "id": 42,
      "username": "user@example.com",
      "email": "user@example.com",
      "display_name": "John Doe",
      "subscription_level": "paid"
    }
  },
  "error": null,
  "meta": { "api_version": "1.0.0", "timestamp": 1739404800 }
}
```

ERROR RESPONSE 401

```
{
  "success": false,
  "data": null,
  "error": {
    "code": "INVALID_CREDENTIALS",
    "message": "Invalid username or password.",
    "details": null
  },
  "meta": { "api_version": "1.0.0", "timestamp": 1739404800 }
}
```

POST /auth/token-exchange

Exchange a WordPress session cookie (from WebView login) for JWT tokens. Use this after the user completes login via the WebView flow in the mobile app.

 Requires WordPress session cookie (sent as `Cookie` header)

REQUEST HEADERS

```
Cookie: wordpress_logged_in_abc123=user%7C1739404800%7Ctoken%7Chmac
```

REQUEST BODY

```
{
  "device_id": "android_abc123",
  "device_name": "Pixel 8 Pro",
  "platform": "android"
}
```

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": {
    "access_token": "eyJhbGciOiJIUzI1NiIs... ",
    "refresh_token": "x9y8z7w6v5u4t3...",
    "token_type": "Bearer",
    "expires_in": 1800,
    "user": {
      "id": 42,
      "username": "user@example.com",
      "email": "user@example.com",
      "display_name": "John Doe",
      "subscription_level": "paid"
    }
  },
  "error": null,
  "meta": { "api_version": "1.0.0", "timestamp": 1739404800 }
}
```

ERROR RESPONSE 401

```
{
  "success": false,
  "data": null,
  "error": {
    "code": "NO_WP_SESSION",
    "message": "No WordPress session cookie found. Log in via WebView first.",
    "details": null
  },
}
```

```
"meta": { ... }  
}
```

POST /auth/refresh

Rotate a refresh token to get new access + refresh tokens. The old refresh token is revoked.

 No Bearer token required; uses refresh token in body

REQUEST BODY

```
{  
  "refresh_token": "a1b2c3d4e5f6g7h8i9j0..."  
}
```

SUCCESS RESPONSE 200

```
{  
  "success": true,  
  "data": {  
    "access_token": "eyJhbGciOiJIUzI1NiIs... (new)",  
    "refresh_token": "q1w2e3r4t5y6u7... (new)",  
    "token_type": "Bearer",  
    "expires_in": 1800  
  },  
  "error": null,  
  "meta": { ... }  
}
```

ERROR RESPONSE 401

```
{  
  "success": false,  
  "data": null,  
  "error": {  
    "code": "REFRESH_FAILED",  
    "message": "Invalid or expired refresh token. Please log in again.",  
    "details": null  
  },  
  "meta": { ... }  
}
```

POST /auth/logout

Revoke a refresh token. Optionally revoke all tokens for the user (logout all devices).

 Requires Bearer token

REQUEST BODY

```
{
  "refresh_token": "a1b2c3d4e5...",
  "all_devices": false
}
```

FIELD	TYPE	REQUIRED	DESCRIPTION
<code>refresh_token</code>	string	No	Specific token to revoke
<code>all_devices</code>	boolean	No	Set <code>true</code> to revoke ALL sessions

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": { "message": "Logged out successfully." },
  "error": null,
  "meta": { ... }
}
```

GET `/auth/verify`

Verify that the current access token is valid. Returns user info and active session count.

 Requires Bearer token

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": {
    "valid": true,
    "user": {
      "id": 42,
      "username": "user@example.com",
      "email": "user@example.com",
      "display_name": "John Doe",
      "subscription_level": "paid"
    },
    "active_sessions": 2
  },
}
```

```
"error": null,  
"meta": { ... }  
}
```

User

GET /user/profile

Get the authenticated user's profile including subscription level and admin status.

 Requires Bearer token

SUCCESS RESPONSE 200

```
{  
  "success": true,  
  "data": {  
    "id": 42,  
    "username": "user@example.com",  
    "email": "user@example.com",  
    "display_name": "John Doe",  
    "subscription_level": "paid",  
    "is_admin": false,  
    "active_sessions": 2  
  },  
  "error": null,  
  "meta": { ... }  
}
```

GET /user/preferences

Get the user's mobile app preferences (sync settings, theme, etc). Returns key-value pairs.

 Requires Bearer token

SUCCESS RESPONSE 200

```
{  
  "success": true,  
  "data": {  
    "auto_sync_enabled": true,  
    "sync_frequency": "DAILY",  
    "sync_on_app_open": true,  
  }  
}
```

```
{
  "background_sync_enabled": true,
  "show_sync_notifications": true,
  "delete_gps_after_sync": false,
  "sync_over_mobile_data": true,
  "distance_unit": "miles",
  "theme": "dark"
},
{
  "error": null,
  "meta": { ... }
}
```

Note: Returns empty object {} if no preferences have been set.

PUT /user/preferences

Update one or more mobile app preferences. Only allowed keys are saved; unknown keys are ignored.

 Requires Bearer token

ALLOWED PREFERENCE KEYS

KEY	TYPE	VALUES
auto_sync_enabled	boolean	true/false
sync_frequency	string	AFTER_TRIP, DAILY, WEEKLY
sync_on_app_open	boolean	true/false
background_sync_enabled	boolean	true/false
show_sync_notifications	boolean	true/false
delete_gps_after_sync	boolean	true/false
sync_over_mobile_data	boolean	true/false
distance_unit	string	miles, km
theme	string	light, dark, system
notification_reminders	boolean	true/false

REQUEST BODY

```
{
  "auto_sync_enabled": true,
  "sync_frequency": "DAILY",
  "theme": "dark"
}
```

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": {
    "updated": {
      "auto_sync_enabled": true,
      "sync_frequency": "DAILY",
      "theme": "dark"
    }
  },
  "error": null,
  "meta": { ... }
}
```

Vehicles

GET /vehicles

List all vehicles belonging to the authenticated user.

 Requires Bearer token

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": [
    {
      "id": "v_1766985747930_q4x1fvngxzde",
      "user_id": "42",
      "name": "My Daily Driver",
      "display_name": "2018 Honda Accord",
      "vin": "1HGBH41JXMN109186",
      "plate": "ABC1234",
      "year": 2018,
      "make": "Honda",
      "model": "Accord",
      "engine": "1.5L Turbo I4",
      "body_class": "Sedan",
      "odometer": 52392,
      "photo_path": "/uploads/vehicles/honda_accord.jpg",
      "insurance_expiry": "2026-06-15",
      "registration_expiry": "2026-03-20"
    },
    {
      "id": "v_1766985812456_m8k2prnxw7h",
      "user_id": "42",
      "name": "Weekend Car",

```

```
    "display_name": "2015 Toyota Camry",
    "vin": null,
    "plate": "XYZ5678",
    "year": 2015,
    "make": "Toyota",
    "model": "Camry",
    "engine": null,
    "body_class": null,
    "odometer": 78900,
    "photo_path": null,
    "insurance_expiry": null,
    "registration_expiry": null
  }
],
"error": null,
"meta": { ... }
}
```

GET /vehicles/{id}

Get a single vehicle by its ID. Returns 404 if not found or not owned by user.

 Requires Bearer token

URL PARAMETERS

PARAM	TYPE	EXAMPLE	DESCRIPTION
{id}	string	v_1766985747930_q4x1fvngxzde	Vehicle ID

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": {
    "id": "v_1766985747930_q4x1fvngxzde",
    "user_id": "42",
    "name": "My Daily Driver",
    "display_name": "2018 Honda Accord",
    "vin": "1HGBH41JXMN109186",
    "plate": "ABC1234",
    "year": 2018,
    "make": "Honda",
    "model": "Accord",
    "engine": "1.5L Turbo I4",
    "body_class": "Sedan",
    "odometer": 52392,
    "photo_path": "/uploads/vehicles/honda_accord.jpg",
    "insurance_expiry": "2026-06-15",
    "registration_expiry": "2026-03-20"
  }
}
```

```
},
"error": null,
"meta": { ... }
}
```

ERROR RESPONSE 404

```
{
  "success": false,
  "data": null,
  "error": {
    "code": "NOT_FOUND",
    "message": "Vehicle not found.",
    "details": null
  },
  "meta": { ... }
}
```

PUT /vehicles/{id}/odometer

Update a vehicle's odometer reading. Cannot decrease. Max single jump: 10,000 miles.

 Requires Bearer token

URL PARAMETERS

PARAM	TYPE	EXAMPLE
{id}	string	v_1766985747930_q4x1fvngxzde

REQUEST BODY

```
{
  "odometer": 52500
}
```

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": {
    "vehicle_id": "v_1766985747930_q4x1fvngxzde",
    "previous_odometer": 52392,
    "new_odometer": 52500,
    "increase": 108
  },
  "error": null,
}
```

```
"meta": { ... }
}
```

ERROR: DECREASE 422

```
{
  "success": false,
  "data": null,
  "error": {
    "code": "ODOMETER_DECREASE",
    "message": "New odometer (50000) cannot be less than current (52392).",
    "details": null
  },
  "meta": { ... }
}
```

ERROR: JUMP TOO LARGE 422

```
{
  "success": false,
  "data": null,
  "error": {
    "code": "ODOMETER_JUMP_TOO_LARGE",
    "message": "Odometer increase of 15000 miles exceeds maximum allowed.",
    "details": null
  },
  "meta": { ... }
}
```

GET /vehicles/{id}/reminders

Get all maintenance reminders for a specific vehicle.

 Requires Bearer token

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": [
    {
      "id": "r_17669857_abc123",
      "vehicle_id": "v_1766985747930_q4x1fvngxzde",
      "vehicle_name": "My Daily Driver",
      "service_name": "Oil Change",
      "title": "Regular oil change",
      "due_date": "2026-03-15",
    }
  ]
}
```

```
    "due_mileage": 55000,
    "current_odometer": 52392,
    "base_odo": 50000,
    "base_date": "2025-09-15",
    "is_recurring": true,
    "interval_miles": 5000,
    "interval_months": 6,
    "notes": "Use synthetic 0W-20",
    "created_at": "2025-09-15 10:00:00",
    "updated_at": "2025-09-15 10:00:00"
  }
],
"error": null,
"meta": { ... }
}
```

GET /vehicles/{id}/reminders/due

Get due or overdue reminders for a specific vehicle. Includes urgency level.

 Requires Bearer token

QUERY PARAMETERS

PARAM	TYPE	DEFAULT	DESCRIPTION
days	integer	30	Look-ahead window in days

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": {
    "vehicle_id": "v_1766985747930_q4x1fvngxzde",
    "window_days": 30,
    "count": 1,
    "reminders": [
      {
        "id": "r_17669857_abc123",
        "vehicle_id": "v_1766985747930_q4x1fvngxzde",
        "vehicle_name": "My Daily Driver",
        "service_name": "Tire Rotation",
        "title": "Rotate tires",
        "due_date": "2026-02-20",
        "due_mileage": 53000,
        "current_odometer": 52392,
        "base_odo": 48000,
        "base_date": "2025-08-20",
        "is_recurring": true,

```

```
    "interval_miles": 5000,
    "interval_months": 6,
    "notes": null,
    "created_at": "2025-08-20 14:30:00",
    "updated_at": "2025-08-20 14:30:00",
    "is_overdue": false,
    "urgency": "upcoming"
  }
]
},
"error": null,
"meta": { ... }
}
```

URGENCY LEVELS

overdue	Past due date OR past due mileage
urgent	Within 7 days or 500 miles
upcoming	Within 30 days or 2,000 miles
normal	Further out

Reminders

GET /reminders

List all maintenance reminders across all user vehicles.

 Requires Bearer token

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": [
    {
      "id": "r_17669857_abc123",
      "vehicle_id": "v_1766985747930_q4x1fvgxzde",
      "vehicle_name": "My Daily Driver",
      "service_name": "Oil Change",
      "title": "Regular oil change",
      "due_date": "2026-03-15",
      "due_mileage": 55000,
      "current_odometer": 52392,

```

```
    "base_odo": 50000,
    "base_date": "2025-09-15",
    "is_recurring": true,
    "interval_miles": 5000,
    "interval_months": 6,
    "notes": "Use synthetic 0W-20",
    "created_at": "2025-09-15 10:00:00",
    "updated_at": "2025-09-15 10:00:00"
  }
],
"error": null,
"meta": { ... }
}
```

GET /reminders/due

Get all due/overdue reminders across all vehicles. Includes urgency and overdue flag.

 Requires Bearer token

QUERY PARAMETERS

PARAM	TYPE	DEFAULT	DESCRIPTION
<code>days</code>	integer	30	Look-ahead window in days

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": {
    "window_days": 30,
    "total": 3,
    "overdue_count": 1,
    "reminders": [
      {
        "id": "r_17669857_def456",
        "vehicle_id": "v_1766985812456_m8k2prnxw7h",
        "vehicle_name": "Weekend Car",
        "service_name": "Brake Inspection",
        "title": "Check brake pads",
        "due_date": "2026-01-10",
        "due_mileage": null,
        "current_odometer": 78900,
        "base_odo": null,
        "base_date": "2025-07-10",
        "is_recurring": true,
        "interval_miles": null,
        "interval_months": 6,

```

```
    "notes": null,
    "created_at": "2025-07-10 09:00:00",
    "updated_at": "2025-07-10 09:00:00",
    "is_overdue": true,
    "urgency": "overdue"
  }
]
},
"error": null,
"meta": { ... }
}
```

GET /reminders/{id}

Get a single reminder by ID.

 Requires Bearer token

URL PARAMETERS

PARAM	TYPE	EXAMPLE
{id}	string	r_17669857_abc123

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": {
    "id": "r_17669857_abc123",
    "vehicle_id": "v_1766985747930_q4x1fvngxzde",
    "vehicle_name": "My Daily Driver",
    "service_name": "Oil Change",
    "title": "Regular oil change",
    "due_date": "2026-03-15",
    "due_mileage": 55000,
    "current_odometer": 52392,
    "base_odo": 50000,
    "base_date": "2025-09-15",
    "is_recurring": true,
    "interval_miles": 5000,
    "interval_months": 6,
    "notes": "Use synthetic 0W-20",
    "created_at": "2025-09-15 10:00:00",
    "updated_at": "2025-09-15 10:00:00"
  },
  "error": null,
}
```

```
"meta": { ... }
}
```

Sync

POST /sync/push

Batch update odometer readings for multiple vehicles. This is the primary sync endpoint used by the mobile app after trip tracking. Sends only odometer deltas — **no GPS data**.

 Requires Bearer token

REQUEST HEADERS (OPTIONAL)

```
X-Device-Id: android_abc123
```

REQUEST BODY

```
{
  "vehicles": [
    {
      "id": "v_1766985747930_q4x1fvngxzde",
      "odometer": 52534
    },
    {
      "id": "v_1766985812456_m8k2prnxw7h",
      "odometer": 78920
    }
  ]
}
```

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": {
    "synced": 2,
    "total": 2,
    "miles": 162,
    "results": [
      {
        "id": "v_1766985747930_q4x1fvngxzde",
        "success": true,
        "previous_odometer": 52392,
        "new_odometer": 52534,

```

```

        "increase": 142
      },
      {
        "id": "v_1766985812456_m8k2prnxw7h",
        "success": true,
        "previous_odometer": 78900,
        "new_odometer": 78920,
        "increase": 20
      }
    ]
  },
  "error": null,
  "meta": { ... }
}

```

PARTIAL SUCCESS RESPONSE 207

```

{
  "success": true,
  "data": {
    "synced": 1,
    "total": 2,
    "miles": 142,
    "results": [
      {
        "id": "v_1766985747930_q4x1fvngxzde",
        "success": true,
        "previous_odometer": 52392,
        "new_odometer": 52534,
        "increase": 142
      },
      {
        "id": "v_invalid_id",
        "success": false,
        "error": "Vehicle not found or not owned by user"
      }
    ]
  },
  "error": null,
  "meta": { ... }
}

```

Note: Status 207 (Multi-Status) is returned when some vehicles succeed and some fail.

GET /sync/status

Get sync history for the current user/device. Pass `X-Device-Id` header for device-specific stats.

 Requires Bearer token

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": {
    "last_sync_push": "2026-02-10 14:30:00",
    "last_sync_pull": null,
    "vehicles_synced": 15,
    "total_miles_synced": 1247.50,
    "sync_count": 8,
    "last_error": null
  },
  "error": null,
  "meta": { ... }
}
```

Note: Returns zeroed values if no sync history exists yet.

POST /sync/register-device

Register a mobile device for push notifications and sync tracking.

 Requires Bearer token

REQUEST BODY

```
{
  "device_id": "android_abc123",
  "platform": "android",
  "device_name": "Pixel 8 Pro",
  "device_model": "Pixel 8 Pro",
  "os_version": "15",
  "app_version": "1.0.0",
  "push_token": "fcm:dRz8x9Yk2T..."
}
```

FIELD	TYPE	REQUIRED	DESCRIPTION
device_id	string	Yes	Unique device identifier
platform	string	Yes	android or ios
device_name	string	No	Friendly name
push_token	string	No	FCM/APNS push notification token

SUCCESS RESPONSE 201

```
{
  "success": true,
  "data": {
    "status": "registered",
    "device_id": "android_abc123"
  },
  "error": null,
  "meta": { ... }
}
```

Subscription

GET `/subscription/status`

Get the user's subscription level and feature flags. Use this to determine which features to enable in the mobile app.

 Requires Bearer token

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": {
    "subscription_level": "paid",
    "features": {
      "manual_sync": true,
      "background_sync": true,
      "auto_sync": true,
      "trip_tracking": true,
      "vehicle_management": true,
      "odometer_adjustment": true,
      "reminders": true
    }
  },
  "error": null,
  "meta": { ... }
}
```

FREE USER RESPONSE

```
{
  "success": true,
  "data": {
    "subscription_level": "free",
    "features": {
      "manual_sync": true,
```

```
    "background_sync": true,
    "auto_sync": false,
    "trip_tracking": true,
    "vehicle_management": true,
    "odometer_adjustment": true,
    "reminders": true
  }
},
"error": null,
"meta": { ... }
}
```

Admin

All admin endpoints require WordPress **administrator** role.

GET /admin/test

Verify admin access is working.

 Requires Bearer token + Admin role

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": {
    "message": "Admin API Tester active.",
    "admin_user": {
      "user_id": 1,
      "username": "admin",
      "email": "admin@trackmywrench.com"
    }
  },
  "error": null,
  "meta": { ... }
}
```

GET /admin/users

List all users with subscription info. Paginated.

 Requires Bearer token + Admin role

QUERY PARAMETERS

PARAM	DEFAULT	DESCRIPTION
page	1	Page number
per_page	50	Results per page (max 100)

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": [
    {
      "id": 1,
      "username": "admin",
      "email": "admin@trackmywrench.com",
      "display_name": "Admin",
      "subscription_level": "paid"
    }
  ],
  "error": null,
  "meta": {
    "api_version": "1.0.0",
    "timestamp": 1739404800,
    "pagination": {
      "total": 25,
      "page": 1,
      "per_page": 50,
      "total_pages": 1,
      "has_more": false
    }
  }
}
```

GET [/admin/logs](#)

View API request logs. Paginated with filters.

 Requires Bearer token + Admin role

QUERY PARAMETERS

PARAM	DESCRIPTION
user_id	Filter by user ID
endpoint	Filter by endpoint path (partial match)
status_code	Filter by HTTP status code

method	Filter by HTTP method
page	Page number (default 1)
per_page	Results per page (max 200, default 50)

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": [
    {
      "id": 1,
      "user_id": 42,
      "method": "GET",
      "endpoint": "/vehicles",
      "status_code": 200,
      "response_time_ms": 45,
      "ip_address": "192.168.1.1",
      "device_id": "android_abc123",
      "error_message": null,
      "created_at": "2026-02-12 14:30:00"
    }
  ],
  "error": null,
  "meta": {
    "pagination": { "total": 150, "page": 1, "per_page": 50, "total_pages": 3, "has_more": true }
  }
}
```

GET /admin/stats

API usage statistics: request counts, auth metrics, device counts, sync totals, top endpoints.

 Requires Bearer token + Admin role

SUCCESS RESPONSE 200

```
{
  "success": true,
  "data": {
    "requests": {
      "total_24h": 342,
      "total_7d": 2150,
      "errors_24h": 12,
      "avg_response_ms": 38.5
    },
    "auth": {
```

```
        "active_tokens": 8,
        "unique_users_24h": 5
    },
    "devices": {
        "total_registered": 12,
        "by_platform": [
            { "platform": "android", "count": 10 },
            { "platform": "ios", "count": 2 }
        ]
    },
    "sync": {
        "total_syncs": 45,
        "total_miles": 3250.75
    },
    "top_endpoints_24h": [
        { "endpoint": "/vehicles", "method": "GET", "count": 89, "avg_ms": 35.2 },
        { "endpoint": "/reminders/due", "method": "GET", "count": 67, "avg_ms": 42.1 }
    ]
},
"error": null,
"meta": { ... }
}
```

Mobile Integration Guide

Android (Kotlin) — Login Flow

```
// Option A: Direct Login
val response = api.login(LoginRequest(
    username = "user@example.com",
    password = "secret",
    device_id = getDeviceId(),
    platform = "android"
))
sessionManager.saveTokens(
    response.access_token,
    response.refresh_token
)

// Option B: WebView → Token Exchange
// 1. User logs in via WebView to /login?mobile_app=1
// 2. On redirect to /app/?login_success=1:
val cookies = CookieManager.getInstance()
    .getCookie("https://trackmywrench.com")
val response = api.tokenExchange(cookies)
sessionManager.saveTokens(...)
```

Token Refresh (OkHttp Interceptor)

```
class AuthInterceptor : Interceptor {
    override fun intercept(chain: Chain): Response {
        val request = chain.request().newBuilder()
            .addHeader("Authorization",
                "Bearer " + sessionManager.accessToken)
            .build()
        val response = chain.proceed(request)
        if (response.code == 401) {
            // Refresh token and retry
            val newTokens = api.refresh(
                sessionManager.refreshToken)
            sessionManager.saveTokens(newTokens)
            // Retry original request with new token
        }
        return response
    }
}
```

Sync Push (After Trip Tracking)

```
// After trips complete, batch update odometers
val syncPayload = SyncPushRequest(
    vehicles = pendingVehicles.map {
        VehicleUpdate(
            id = it.id, // String ID!
            odometer = it.adjustedOdometer
                ?: it.calculatedOdometer
        )
    }
)
val result = api.syncPush(syncPayload)
// Mark successful vehicles as synced
result.results
    .filter { it.success }
    .forEach { markVehicleSynced(it.id) }
```

Privacy

GPS data **never leaves the device**. Only odometer deltas are synced. The mobile app calculates distance locally using GPS, then sends only the final odometer number per vehicle.



Security

TLS: All traffic enforced HTTPS via .htaccess.

JWT: HMAC-SHA256. Access: 30 min. Refresh: 30 days, rotated on use, stored as SHA-256 hash.

Rate Limiting: Database-backed sliding window per IP and per user.

Certificate Pinning: Recommended for mobile apps.

Input Validation: All inputs sanitized. SQL injection prevented via PDO prepared statements.

CORS: Restricted to `trackmywrench.com` . Mobile apps (no Origin header) pass through.

 Database Tables

API-specific tables (new, do not touch existing GarageMinder tables):

<code>api_refresh_tokens</code>	JWT refresh token storage with device info
<code>api_rate_limits</code>	Request throttling counters
<code>api_devices</code>	Registered mobile devices
<code>api_request_log</code>	Audit trail of all API requests
<code>api_user_preferences</code>	Mobile app settings (key-value)
<code>api_sync_status</code>	Sync tracking per user/device

Existing GarageMinder tables read by API (not modified except `vehicles.current_odo` on sync push):

<code>vehicles</code>	Vehicle data (<code>current_odo</code> updated on sync)
<code>reminders</code>	Maintenance reminders (read-only)